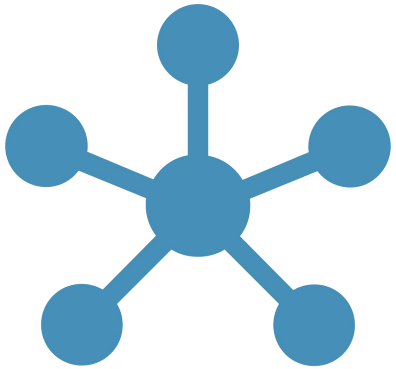
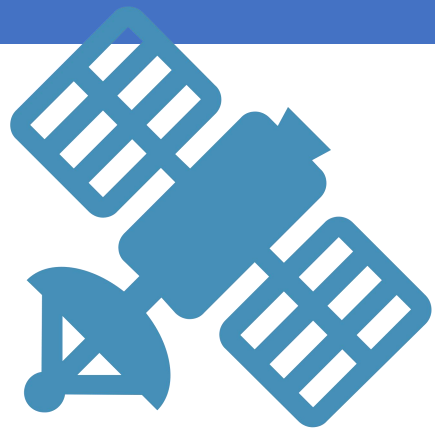
Job Interview Presentation

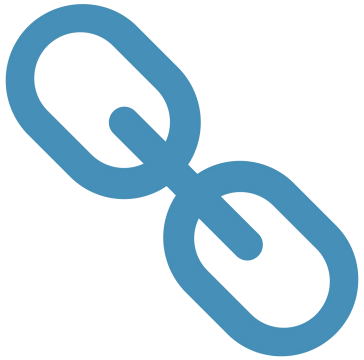
Victoria Lee

The interview will focus on networking fundamentals and consist of 10 questions.





Networking



Devices

Protocols

United States Cyber Command uses state-of-the art technology. However, the fundamentals are as important as any innovative technology. Given that, what are the three basic local area network (LAN) topologies called?

- Bus Topology
- Star Topology
- Ring Topology



The national mission teams operate on all types of networks. To defend a network, you must first understand the network design on which you are operating. Provide a brief description of each of the following network topologies:

Topology	Description
Bus	A straight-line cable that all devices are networked into that cable. (T connector)
Star	A central point from which all attached devices radiate and connected to point-to-point.
Ring	Data is in sent in a single direction of a ring to the connected devices, until the intended destination receives the data.
Mesh	An optimal route between sites, whiles avoiding the interconnecting every site to every other site (Full(star in pentagon)/Partial (arrow in square))
Hybrid	Combines more than two types of network topologies of the same network and prioritizes which path is the fastest case.

Our employees must have a theoretical and applicable understanding of how networking works. Name each layer of the OSI model. Provide the layer number and name from top to bottom.

Layer	OSI Layer Name
Level 7	Application
Level 6	Presentation
Level 5	Session
Level 4	Transport
Level 3	Network
Level 2	Data Link
Level 1	Physical

Briefly describe each function of the OSI model layer. Provide the layer name and the function in your response.



Layer	Function
Application (Layer 7)	Applications allow control over the environment; to configure and monitor it in the network manager.
Presentation (Layer 6)	Formats data being exchanged and encrypts it for transmission across a network; securing data / data security
Session (Layer 5)	This sets up, maintain, and tears down sessions.
Transport (Layer 4)	It is between upper and lower layers; Messages from are encapsulated into segments for transmission in; Data streams are decapsulated and sent to Layer 5
Network (Layer 3)	Forwarding data based on logical addresses, switching, route discovery, and selection
Data Link (Layer 2)	Packaging data into frames; Transmitting it to a network; Error detection/correction; Finding network devices with an address; Handling flow control
Physical (Layer 1)	Transmission of bits on a network along with physical and electrical characteristics such as bits of 1 or 0 in binary or wiring.

United States Cyber Command requires an internet service provider (ISP) to connect to the internet. What is the point at which the operational control of ownership changes from the ISP to United States Cyber Command?



Demarcation Point

Our teams operate and encounter all types of devices. Provide a brief description of each of the following common network devices:

Networking Device	Description
Router	It connects different networks together for routing data based on IP address.
Hub	It allows multiple devices or networks to be connected to broadcast data.
Switch	It is a device that connects other devices on a network using address, port, and helps move the traffic (data) for the best scenario and destination.
NIC (Network Interface Card)	It is a hardware part of a computer so that it allows the ability to connect to a network.

The national mission teams require implementation of common protocols. Provide the port that each of the following protocols use:

Protocol	Port
HTTP	TCP 80
SMTP	TCP 25
SNMP	UDP 161, 162
DNS	TCP/UDP 53
HTTPS	TCP 443
DHCP	UDP 67, 68
TELNET	TCP 23

Provide the IP range for each network class:



- Class A: 0.0.0.0 - 127.0.0.0
- Class B: 128.0.0.0 - 191.255.0.0
- Class C: 192.0.0.0 - 223.255.255.0

Note: The question is asking for the class range, not the private IP range for each class.

Provide the default subnet mask to the class of network:



Class	Format	Default Subnet Mask
A	network.host.host.host	255.0.0.0
B	network.network.host.host	255.255.0.0
C	network.network.network.host	255.255.255.0

Describe the purpose of an autonomous system.



- **Purpose:** one or more network operators that control a group a different UP networks ad routers
- **Routing policies:** It allows routing policies independently
- **Internet Routing:** It allows the internet to be as a network of interconnected networks
- **BGP (Border Gateway Protocol):** It allows BGP to exchange routing information with other autonomous systems for global internet infrastructure
- **Network Autonomy:** It allows each system to decide which networks to route traffic to and from with other autonomous systems

Thank You!

Upload your PowerPoint and video to the Assignments folder to receive a grade and feedback.

